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Response to “Comment on ‘Simultaneous density and thermal conductivity depth profile reconstructions from noised thermal-wave amplitude and phase data using a combined integral equation and imperialist competitive algorithm method’” [J. Appl. Phys. 134, 056101 (2023)]

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Right at the outset, we wish to make clear that we *do not dispute* the validity of the Liouville transformation approach proposed in the Comment¹ to our paper,² which leads to the emergence of the non-uniqueness issues of the reconstructed depth profiles presented in it. Nevertheless, the Liouville transformation (or normal form) is only *one way* of converting the Sturm–Liouville equation (here, the frequency-domain thermal-wave equation) and the associated boundary-value problem (BVP), which groups together the generalized thermophysical parameters under the definition of $\xi(x)$ in Eq. (2) of the Comment. It is well known³ that all contributing thermophysical properties, conductivity and volumetric heat capacity (VHC; here, density), vary in a non-identifiable manner within the $\xi(x)$ integrand of Eq. (2) and in the thermal-wave BVP of Eqs. 3(a) and 3(b) in a way that the universal $b(\xi)$ coefficient lumps together and inseparably these contributions. Unfortunately, this normal transformation has the disadvantage of enhancing non-identifiability of individual thermophysical properties that together constitute the generalized inverse thermal diffusivity, Eq. (2), and resulting thermal effusivity depth profiles, Eqs. 3

(a) and 3(b). For arbitrarily depth-variable properties, the transformation reduces the thermal-wave BVP to the unique depth-variable effusivity parameter. As a result, this generates the effusivity vs $\xi^{1/2}$ dependence plotted in Fig. 1 of the Comment, which *renormalizes the depth scale* (a standard feature of normal or normalizing transformations) while rendering individual contributor parameters—thermal conductivity and VHC—inseparable and, therefore, individually unidentifiable. The single $b(\xi)$ ($\xi(x)$) depth profile reduces all individual depth variations to their product (ratio), thereby hiding individual variations in experimentally derived curves like the one used in Fig. 1, which lead to the demonstrably non-unique results graphed in Fig. 2 and Figs. S1–S4. This is the essence of the non-uniqueness argument that was raised in detail in the Comment.

However, the Liouville transformation is *not the only kind of coordinate transformation* that can be imposed on the Sturm–Liouville equation, especially if parameter uniqueness under depth-variable conditions is at stake. It should be observed that as a result of the Liouville transformation, the single depth-dependent

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parameter $b(\xi)$ appears both as a multiplication factor on the lhs and inside the spatial coordinate derivative on the rhs of Eq. (3a), thereby excluding any possibility of separate identifiability among its constituent parameters. For this reason, this is not the route to transform the original thermal-wave BVP as its main handicap is that it fails to recognize the fact that in the BVP definition, Eqs. 1(a) and 1(b) in the Comment and also Eqs. 1(a) and 1(b) in our paper,² *there is a fundamentally different and asymmetric rate of depth variation between conductivity and density depth profiles* (VHC, with specific heat being assumed a constant here): The conductivity variation appears under the spatial derivative operator d/dx whereas the density enters as a multiplicative factor in the $C(x)$ definition. This observation of asymmetric weights within the thermal-wave ODE, Eq. (1a) in the Comment and Eq. (6) of Ref. 2 is the foundation of our development of an altogether different transformation of the BVP, leading in a novel Fredholm-type integral equation transformation approach to the inverse problem, which is *directly opposite to the Liouville transformation* and aimed at enhancing the *individuality* of the two independent (conductivity and density) asymmetric depth profiles. To accomplish this, a Green function was adopted which only included the conductivity part of the thermal-wave ODE

$$\frac{d}{dx} \left[k(x) \frac{dG(x|x_0)}{dx} \right] = -\delta(x - x_0), \quad (1)$$

while the multiplicative density-dependent term $i\omega C(x)X(x; \omega)$ was entered as a source or perturbation contribution, Eq. (7) of Ref. 2. In this manner, the asymmetry in rates of depth variation between the two thermophysical parameters $k(x)$ and $C(x)$ can be optimally amplified (not parts of the same Green function), paving the way for the identification of each of the parameters from the differences in depth dependencies upon formulating an appropriate integral equation. This is an ideal scenario for using the integral equation method, as the resulting thermal-wave expression involves the parameter $C(x)$ uncoupled from $k(x)$: both parameters appear under a full-depth-range integral, but the latter is weighted under an additional integration from a local coordinate point x_0 to infinity due to the presence of the d/dx operator in Eq. (1). The surface value of the thermal-wave field is thus obtained as Eq. (10) of Ref. 2,

$$T(0; \omega) = -i\omega \int_0^\infty C(x_0) \left[\int_{x_0}^\infty \frac{dy}{k(y)} \right] X(x_0; \omega) dx_0 - k(0) \left(\frac{dX(x_0; \omega)}{dx_0} \Big|_{x_0=0} \right) \int_0^\infty \frac{dy}{k(y)}. \quad (2)$$

This expression clearly shows that under the particular integral equation transformation presented in our paper, the surface value of the thermal-wave field $T(0; \omega)$ does depend on the depth-integrated *distinctly separate* variation profiles of $C(x)$ and $k(x)$ and can be used judiciously to separate out the two contributions with a suitable separation methodology. Notice that if $1/(k(y))$ in Eq. (2) were not in the form of an integral, $\int_{x_0}^\infty \frac{dy}{k(y)}$, inside the brackets, the integrand in Eq. (2) would coalesce into $C(x_0)/k(x_0)$ which is equal to $1/\alpha(x_0)$, the depth-variable thermal

diffusivity which renders the two component functions C and k spatially non-identifiable as in the Liouville transformation.

In seeking the maximum sensitivity to, and reliability of, the two depth profiles, we found that a differential approach between the foregoing fully depth-variable BVP solution converging to (known) bulk values of the thermophysical properties as an additional *a priori* piece of information and the analytical frequency dependent solution of the associated thermal-wave BVP with constant parameters representing the bulk values in a semi-infinite material/region yields an optimal condition for uniquely and sensitively deriving the depth profiles under consideration when used with an optimized computational method such as the imperialist competitive algorithm (ICA) introduced in our work. To accomplish this task, an approximate solution to the integral equation (2) was sought replacing the spatial term $X(x; \omega)$ in the integrand with the well-known solution to the constant-bulk-parameter thermal-wave function

$$X_b(x_0; \omega) = X_b(x_0; p) = T_b(x_0; p) = \frac{F_0}{k_b p} e^{-px_0}; \quad (3)$$

$$p = (i\omega/\alpha_b)^{1/2},$$

with (α_b, k_b) representing the bulk thermal diffusivity and conductivity, respectively. This type of trial solution to an associated simpler parameter problem, a feature of integral equations, is known to yield very good approximations to the desired solution of more complicated profiles in Sturm-Liouville type of problems.⁴ Furthermore, a differential approach further strengthens the uniqueness and reliability of the solutions as it imposes depth-variable normalization bounds. In our theory, the arbitrary conductivity and density-specific heat depth profiles were modelled as depth function variations away from the respective baseline bulk values,

$$k(x) = k_b - \Delta k(x), \quad (4)$$

$$C(x) = C_b - \Delta C(x).$$

The final expressions stemming from Eq. (2) (surface boundary condition terms neglected in this discussion for simplicity) yield differential surface thermal-wave signals between the variable and constant depth profiles in terms of spatial Laplace transforms,

$$\Delta T(0; \omega) \equiv T(0; p) - T_b(0; p) = -\frac{p}{k_b^2} \sum_{m=0}^\infty \frac{1}{k_b^m} \hat{K}_{m+1}(p) + \frac{p}{k_b C_b} \sum_{m=1}^\infty \frac{1}{k_b^m} \hat{H}_m(p), \quad (5a)$$

$$\hat{K}_{m+1}(p) = \int_0^\infty K_{m+1}(x_0) e^{-px_0} dx_0, \quad K_{m+1}(x_0) \equiv \int_{x_0}^\infty \Delta k^{m+1}(y) dy, \quad (5b)$$

$$\hat{H}_m(p) = \int_0^\infty K_m(x_0) \Delta C(x_0) e^{-px_0} dx_0. \quad (5c)$$

Equation (5a) demonstrates the complete separation between the purely $k(x)$ —dependent terms with Laplace transform Eq. (5b),

and *mixed* $C(x)$ and $k(x)$ transform terms Eq. (5c). Unlike the Liouville transformation, the developed integral equation transformation highlights the separability and identifiability of the two parameter depth profiles resulting from their different depth variation rates. The structure of Eqs. (5a)–(5c) stands in juxtaposition to the Liouville transformation property of lumping all depth variations into the same surface temperature as discussed in the Comment. It has the key advantage of yielding *distinct depth integrals* of differential parameters away from their bulk values, the latter acting as known continuous thermal-wave distribution pathways bounding the sought depth profiles within the measured surface temperature excursion from the homogeneous solid values at each modulation frequency.

Equation (5a) shows that the surface thermal-wave field value differs among materials of different depth-variable properties, unlike the universal expression obtained from the Liouville transformation as mentioned in the Comment. Eqs. (5b) and (5c) evoke an iterative scheme of computation where $k(x)$ trial functions can be introduced in the Laplace transformation of Eq. (5b), with the approximate values then implemented as seed values in the integrand of the Laplace transform (5c) to optimize $\Delta C(x)$, followed by the next iteration until the imposed minimum error conditions are met. This approach is well rooted in traditional numerical computation iteration methods such as the Gauss–Seidel method.⁵ For this purpose, the introduction of the global ICA was crucial for efficient computation and assurance of *global error minimizations* leading to dual parameter depth profile reconstructions. The ultimate goal of this study was to show experimentally the ability of the presented integral equation method in combination with the ICA to reconstruct optimal depth profiles of the two thermophysical parameters in an inhomogeneous solid as *distinct unique quantities* in the differential expression of Eqs. (5a)–(5c). With regard to the identifiability issue, it is true that the ill-conditioned nature of diffusion problems such as this one renders the determinant of the ICA sensitivity matrix small and the condition number large. However, we firmly maintain that using suitable inversion techniques can result in the convergence of the inverse algorithm.^{6–15} Specifically, in this study, we used the Moore–Penrose pseudoinverse technique, which is a well-known method for the calculation of the inverse matrix in cases involving small determinants like ours.

In conclusion, the combined integral equation approach and two-stage inverse algorithm in this study were shown to highlight the distinct contributions of two thermophysical property depth

profiles as opposed to the Liouville transformation that lumps those properties together into one parameter. The iterative nature of the theoretical expression for the surface thermal-wave value (amplitude and phase) was optimally coupled with the ICA, which used all the available information to identify *among global weighted searches* the actual thermal conductivity and density in an inhomogeneous metal powder compact sample under the aforementioned conditions and initial parameter value guesses. The identified parameters resulted in our accurate reconstructions of a wide range of density and thermal conductivity depth profiles under simulated conditions, both as single and double reconstructed depth profiles, Figs. 5 and 6 of Ref. 2, and of experimentally driven profiles in our sample, which were subsequently verified through destructive analysis, Figs. 7–9 of Ref. 2. Finally, thermal conductivity depth profiles of solids reported by other authors¹⁶ who used independent single parameter reconstruction algorithms were tested using the ICA algorithm, with our reconstruction of the published depth profile data exhibiting very high accuracy shown in Figs. 5(a) and 5(b) of Ref. 2.

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